TITLE OF THE DISSERTATION IN ALL CAPITAL LETTERS

by

Candidate Full Name

DISSERTATION

Submitted in partial satisfaction of the requirements for degree of

DOCTOR OF PHILOSOPHY

in

Biomedical Sciences

in the

GRADUATE DIVISION

of the

UNIVERSITY OF CALIFORNIA, SAN FRANCISCO

Approved:

Dissertation Chair	Date
Committee Member	Date

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by

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DEDICATION

Dedicated to my mentors, colleagues, family, and friends who supported this journey.

ACKNOWLEDGEMENTS

I express my deepest gratitude to my advisor, dissertation committee members, and laboratory colleagues.

CONTRIBUTIONS

Descriptions of student and co-author contributions for adapted manuscript chapters.

EPIGRAPH

"Equipped with his five senses, man explores the universe around him and calls the adventure Science."

— Edwin Powell Hubble (1948)

ABSTRACT

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A concise synthesis of the research problem, methodology, findings, and conclusions. The abstract must be double-spaced and contain no abbreviations in the title.

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LIST OF ABBREVIATIONS

DNA Deoxyribonucleic Acid

RNA Ribonucleic Acid

UCSF University of California, San Francisco

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Biological research in the modern era relies increasingly on multi-omics profiling and computational data modeling. In this chapter, we outline the foundational principles and state-of-the-art literature motivating this dissertation [1].

1.2 Heading Hierarchy Example

As required by the institutional formatting rubric (Sec III.J), headings must strictly adhere to the four-level hierarchy:

1. **Primary Headings** (`\chapter{}`): ALL CAPITAL LETTERS, bold, centered, double spaced above major sections.
2. **Secondary Headings** (`\section{}`): Sentence Case, bold, left-aligned.
3. **Tertiary Headings** (`\subsection{}`): Sentence Case, italicized, unbolded, left-aligned.
4. **Quaternary Headings** (`\subsubsection{}`): Sentence Case, italicized, in-line with paragraph, followed by a period.

1.2.1 Tertiary Subsection Heading Example

Tertiary headings are italicized and unbolded in sentence case. Paragraph text begins double-spaced underneath.

Quaternary Heading Example. Quaternary headings run directly in line with paragraph text, followed immediately by a period.

1.3 Table Formatting Example

Per Rubric Sec III.A, table titles/captions must be placed **above** the table, single-spaced, with bolded table references.

Table 1.1: Demographic summary of study cohorts evaluated across chapters.

Cohort Name	Ancestry	Sample Size (N)	Platform
Discovery Cohort	Multi-Ancestry	621	EPIC BeadChip
Replication Cohort A	African American	1,394	450K BeadChip
Replication Cohort B	European	729	450K BeadChip

1.4 Figure Formatting Example

Per Rubric Sec III.B, figure titles and legends must be placed **below** the figure and single-spaced.



Figure 1.1: Schematic overview of the analytical study workflow. High-throughput DNA methylation and genomic variant data are preprocessed, followed by ancestry-stratified evaluation of epigenetic clock predictions and methylation quantitative trait loci (meQTL) mapping.

1.5 Block Quotation Formatting Example

Per Rubric Sec I.F and III.C, prose quotations over 4 lines must be single-spaced and indented uniformly:

> Biological aging is a complex, multifaceted process characterized by the gradual deterioration of physiological function and increased susceptibility to disease. Epigenetic modifications, particularly DNA methylation at cytosine-phosphate-guanine dinucleotides, provide a dynamic and quantifiable record of cellular chronological and physiological history.

Double-spaced prose continues here immediately following the quotation block.

REFERENCES

[1] Steve Horvath. “DNA methylation age of human tissues and cell types”. In: *Genome Biology* 14.10 (2013), p. 3156. DOI: 10.1186/gb-2013-14-10-r115.

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